



İzmir Institute of Technology

CHE 311

Heat and Mass Transfer

Fall 2025

TERM PROJECT

FINAL REPORT

GROUP NO: 7

Sena UÇUR-300202015

Fatma Ceren UYGUR-300202029

Esra Nur YILMAZ-300202046

Secem KARAKAYA-300202053

Date: 10.01.2026

İzmir, 2026

ABSTRACT

This study presents the thermal design of a plate heat exchanger (PHE) capable of meeting the 2 MW design heat load for a geothermal district heating application. While the district heating water is heated from 40°C to 65°C, the geothermal fluid enters the heat exchanger at 90°C, and the outlet temperature is determined by the energy balance. The heat exchanger is modelled as a single-pass, counter-flow PHE; sizing is performed using the ϵ -NTU method, taking into account the convection and fouling contributions on both sides of the total thermal resistance.

The effect of the number of plates in the parametric study was investigated for $N_p=80,100,120$. Under the design conditions, the required efficiency $\epsilon = 0.584$ was found, and the corresponding required conductivity was $UA \approx 87.6$ kW/K. For the representative case $N_p=100$, $U \approx 2036$ W/m²·K, $A \approx 43.5$ m², and $UA \approx 88.6$ kW/K were obtained, and it stood out as the lowest plate number achieving the target among the tested values. The geothermal outlet temperature was calculated to be approximately 60.8°C with the selected geothermal flow rate. In general, the two-circuit (brine-clean water) structure reduces the risk of corrosion/contamination by isolating the mineral-containing geothermal fluid from the distribution line, thereby increasing long-term operational reliability and contributing to sustainable operation.

TABLE OF CONTENTS

ABSTRACT	2
TABLE OF CONTENTS	3
LIST OF FIGURES.....	5
LIST OF TABLES	6
NOMENCLATURE	7
1.INTRODUCTION.....	9
2. THEORY	10
2.1. Energy Balance and Heat Duty	10
2.2. Heat Capacity Rate.....	10
2.3. Effectiveness (ε) Framework.....	10
2.4. NTU–Effectiveness Relation (Counter-Current).....	10
2.5. Overall Heat Transfer Coefficient (U)	11
2.5.1. Flow Geometry and Channel Definitions.....	11
2.5.2. Flow Velocity, Reynolds Number and Prandtl Number	11
2.5.3. Nusselt Number Correlation for Plate Heat Exchanger	11
2.5.4. Convective Heat Transfer Coefficient.....	12
2.5.5. Area Calculation.....	12
2.6. Pressure Drop Calculation.....	12
3.DESIGN SPECIFICS	13
3.1 Heat Exchanger Type and Configuration Selection	13
3.2 Design Basis and Temperature at Which Characteristics Are Evaluated	13
3.3 The Effect of Number of Plates (N_p)	13
4.DESIGN AND CALCULATION	15
4.1. Mass Flow Rate Calculations (Given Heat Duty).....	15
4.2. Energy Balance Calculation	16

4.3. Capacity Ratio Terms: <i>C_{min}</i> , <i>C_{max}</i> , <i>Cr</i>	16
4.4. Maximum Possible Heat Transfer and Required Effectiveness	17
4.5. ϵ -NTU Analysis and Target UA	17
4.6. Calculation of Determined UA.....	18
4.6.1. Channel Count and Flow Per Channel	18
4.6.2. Reynolds and Prandtl Numbers.....	18
4.6.3. Convection Coefficient	19
4.6.4. Overall Heat-Transfer Coefficient U.....	20
4.6.5. Area Calculation for Each <i>N_p</i> Value	21
4.6.6. Overall Conductance Calculation <i>UA</i> for <i>N_p</i> Value	21
4.7. Pressure Drop Calculations	22
5. FINAL DESIGN	25
6. RESULTS AND DISCUSSION	26
7. CONCLUSION	29
PROJECT PLANNING.....	30
REFERENCES.....	31

LIST OF FIGURES

<u>Figure</u>	<u>Page</u>
Figure 1: Schematic representation of the internal flow arrangement of the counter-current single-pass plate heat exchanger used in the geothermal district heating system	25

LIST OF TABLES

<u>Table</u>	<u>Page</u>
Table 1: Parameters for mc calculations	16
Table 2: Geothermal Brine Outlet Temperatures	16
Table 3: Ch and Cc Values	17
Table 4: u , Aflow, Ac Values According to the Np Parameter	18
Table 5: Reynolds Number Calculations	19
Table 6: Pr Numbers	19
Table 7: Convection Coefficient and Nusselt Number Calculations	20
Table 8: Fouling Factor According to Fluid Type	20
Table 9: Value of Area and Overall Convection Coefficient	21
Table 10: Aplate Value	21
Table 11: UA Calculations	22
Table 12: Pressure Drop Calculations	24
Table 13: Gannt Chart	30

NOMENCLATURE

A	:Total heat transfer area, m^2
A_{plate}	:Effective heat transfer area of a single plate, m^2
A_{flow}	:Total flow cross-sectional area, m^2
A_c	:Flow cross-sectional area of a single channel, m^2
b	:Channel (plate) width, m
s	:Channel opening (plate spacing, gap), m
C	:Heat capacity rate ($\dot{m} \cdot c_p$), kW/K
C_h, C_c	:Heat capacity rates of the hot and cold fluids, kW/K
c_p	:Specific heat capacity, $\text{kJ}/(\text{kg} \cdot \text{K})$
ε	:Heat exchanger effectiveness, $-$
f	:Friction factor (dependent on plate geometry), $-$
h	:Convective heat transfer coefficient, $\text{W}/(\text{m}^2 \cdot \text{K})$
k	:Thermal conductivity, $\text{W}/(\text{m} \cdot \text{K})$
L	:Effective plate length (heat transfer length), m
$\dot{m}$	:Mass flow rate, kg/s
N_p	:Number of plates
N_{ch}	:Number of parallel channels on the relevant side
NTU	:Number of Transfer Units
Nu	:Nusselt number
Pr	:Prandtl number
Q	:Heat transfer rate (heat duty), kW
Re	:Reynolds number
$R_{f,h}, R_{f,c}$	:Fouling resistances of the hot and cold sides, $\text{m}^2 \cdot \text{K/W}$

ρ	:Fluid density, kg/m ³
U	:Overall heat transfer coefficient, W/(m ² ·K)
UA	:Overall thermal conductance, W/K
u	:Average channel flow velocity, m/s
ΔP	:Pressure drop, Pa
h	:Hot fluid (geothermal brine)
c	:Cold fluid (district heating water)
in	:Inlet
out	:Outlet
ΔP_p	:Pressure drop across plate channel, Pa
j_f	:Friction factor (dependent on plate geometry)
L_p	:Flow path length, m
D_h	:Hydraulic diameter, m
ΔP_{pt}	:Port pressure loss, Pa
u_{pt}	:Intraport speed, m/s
D_{pt}	:Port diameter
N_{pass}	:Number of pass
$T_{h,in}$	:Inlet temperature of geothermal brine, °C
$T_{h,out}$	:Outlet temperature of geothermal brine, °C
$T_{c,in}$	:Inlet temperature of district heating water, °C
$T_{c,out}$	:Outlet temperature of district heating water, °C

1.INTRODUCTION

Geothermal district heating systems are important among renewable energy applications, particularly in urban areas, because they provide a long-term, environmentally benign, and continuous heat supply. Thermal energy derived from subsurface geothermal reservoirs is gathered centrally and delivered to various user groups, including residential, commercial, and public facilities. When compared to fossil fuel-based heating systems, geothermal district heating systems have reduced carbon emissions and a more steady energy source, making them a long-term, sustainable solution.

However, geothermal fluids contain dissolved minerals, salts, and other chemical components. This restricts the direct use of geothermal fluids in distribution networks due to corrosion, fouling, and increased maintenance requirements. To address these operational issues, geothermal energy is typically transmitted to a clean secondary water circuit using a heat exchanger. This method improves system reliability while increasing the life of the distribution line.

This research focuses on the design of a heat exchanger capable of handling an approximate thermal load of 2 MW in a low-temperature geothermal district heating system. During the design phase, it was anticipated that the geothermal fluid enters the system at an inlet temperature of 90 °C and a certain mass flow rate; the district heating water must be heated between an entrance temperature of 40 °C and an output temperature of 65 °C. In contrast, the geothermal fluid output temperature was not predetermined; it was established by calculations based on heat exchanger sizing and layout. This method provides a more realistic and engineering-consistent evaluation of system performance.

The long-term operational requirements of geothermal systems were considered when deciding on a heat exchanger type. Within this framework, a counter-flow, single-pass plate heat exchanger was chosen as a good alternative for this study because to its high heat transfer coefficient, compact structure, little space demand, and ease of maintenance. The efficiency-NTU approach was used to analyze the performance of the chosen heat exchanger in the final report. The main design parameters were specified, and the effect of the number of plates on system performance was explored using a parametric analysis. Based on the findings, a final heat exchanger design was provided that fits both the desired thermal load and the operational limits.

2. THEORY

2.1. Energy Balance and Heat Duty

For a steady-flow heat exchanger operating at steady state and neglecting kinetic/potential energy changes and heat losses to the surroundings, the heat transfer rate is obtained from an energy balance on each stream:

$$Q = \dot{m}_h \cdot c_{p,h} \cdot (T_{h,i} - T_{h,o}) \quad (1)$$

$$Q = \dot{m}_c \cdot c_{p,c} \cdot (T_{c,o} - T_{c,i}) \quad (2)$$

2.2. Heat Capacity Rate

The heat capacity ratio C represents the amount of heat a fluid can carry per unit temperature change. This quantity is one of the fundamental parameters determining the thermal behaviour of the fluid in the heat exchanger. A fluid with a higher C value is more resistant to temperature changes.

$$(\dot{m} \cdot C_p)_h = C_h \quad (3)$$

$$(\dot{m} \cdot C_p)_c = C_c \quad (4)$$

2.3. Effectiveness (ε) Framework

This equation represents the theoretically possible maximum heat transfer. This situation occurs when the fluid with the lower heat capacity ratio heats up or cools down to the inlet temperature of the counter-flowing fluid. In real systems, this value is generally unattainable, but it is used as a reference for performance evaluation.

$$Q_{\max} = C_{\min} \times (T_{h,i} - T_{c,i}) \quad (5)$$

Effectiveness is the ratio of actual heat transfer to theoretical maximum heat transfer. This dimensionless parameter indicates how efficiently the heat exchanger is operating and allows for the comparison of different designs.

$$\varepsilon = \frac{Q}{Q_{\max}} \quad (6)$$

$$C_r = \frac{C_{\min}}{C_{\max}} \quad (7)$$

2.4. NTU–Effectiveness Relation (Counter-Current)

NTU represents the dimensionless size of the heat exchanger and incorporates the effect of the heat transfer area and the total heat transfer coefficient. High NTU values indicate a larger surface area or a higher heat transfer coefficient and generally provide higher efficiency.

$$NTU = \frac{U.A}{C_{min}} \quad (8)$$

$$NTU = \frac{1}{C_r - 1} \ln \left(\frac{\varepsilon - 1}{\varepsilon C_r - 1} \right) \quad \text{for } C_r < 1 \quad (9)$$

2.5. Overall Heat Transfer Coefficient (U)

This equation expresses $\frac{1}{UA}$ as the sum of the hot- and cold-side convective resistances, fouling resistances, and the wall conduction resistance. The convective coefficients h_{hh_hhh} and h_{ch_chc} are obtained from correlations, and the resulting UA is used directly for ε -NTU sizing.

$$\frac{1}{UA} = \frac{1}{h_h \cdot A_h} + \frac{R_{f,h}}{A_h} + R_w + \frac{R_{f,c}}{A_c} + \frac{1}{h_c \cdot A_c} \quad (10)$$

$$R_w = \frac{t}{k \cdot A_w} \quad (11)$$

2.5.1. Flow Geometry and Channel Definitions

$$D_h = \frac{4A_c}{P_w} \quad (12)$$

$$A_c = b \cdot s \quad (13)$$

$$P_w = 2(b + s) \quad (14)$$

2.5.2. Flow Velocity, Reynolds Number and Prandtl Number

$$Re = \frac{\rho \cdot u \cdot D_h}{\mu} \quad (15)$$

$$u = \frac{\dot{m}}{\rho \cdot A_{flow}} \quad (16)$$

$$A_{flow} = N_{ch} \cdot A_c \quad (17)$$

$$Pr = \frac{c_p \cdot \mu}{k} \quad (18)$$

2.5.3. Nusselt Number Correlation for Plate Heat Exchanger

$$Nu = \frac{h \cdot D_h}{k} \quad (19)$$

$$[Nu = C_1 Re^p Pr^{C_3}] \quad (20)$$

with $C_1 = 0.2414$, $p = 0.64$, $C_3 = 0.4$

if

$$\left[\begin{array}{c} 700 < Re < 20000 \\ \beta = 45^\circ \end{array} \right]$$

$$Nu = 0.2414 Re^{0.64} Pr^{0.4} \quad (21)$$

2.5.4. Convective Heat Transfer Coefficient

$$h = \frac{Nu * k}{D_h} \quad (22)$$

2.5.5. Area Calculation

$$A = N_p A_{plate} \quad (23)$$

2.6. Pressure Drop Calculation

$$\Delta P_p = 8 j_f \left(\frac{L_p}{D_h} \right) \left(\frac{\rho u^2}{2} \right) \quad (24)$$

$$L_p = L * \text{pass number} \quad (25)$$

$$j_f = 0.6 * Re^{-0.3} \quad (26)$$

$$\Delta P_{pt} = 1.3 \left(\frac{\rho u_{pt}^2}{2} \right) * N_p \quad (27)$$

$$u_{pt} = \frac{\dot{m}}{\rho * A_p} \quad (28)$$

$$A_p = \frac{\pi D_{pt}^2}{4} \quad (29)$$

$$u_{pt} = \frac{4 \dot{m}}{\rho \pi D_{pt}^2} \quad (30)$$

$$\Delta P_{total} = \Delta P_p + \Delta P_{pt} \quad (31)$$

3.DESIGN SPECIFICS

3.1 Heat Exchanger Type and Configuration Selection

For this project, an indirect (two-circuit) system was chosen to avoid mixing the district heating water with the geothermal brine. This increases operational reliability by preventing the transfer of scale and corrosion effects brought on by dissolved minerals in the geothermal fluid to the network. The type of heat exchanger that was selected was a gasketed PHE (Plate Heat Exchanger), which has a compact design, good heat transfer efficiency in liquid-to-liquid applications, and the ability to be opened and cleaned if contaminated.

In order to improve the effective temperature differential, the heat exchanger has been chosen to work in single-pass (1–1) mode and counter-current configuration. The gasket material has been designated as EPDM (Ethylene Propylene Diene Monomer). The preliminary design process was guided by the typical pressure drop levels for plate heat exchangers reported in the literature during the project's interim phase. As indicative upper limit values commonly found in practice, levels of roughly $\Delta P_h \leq 50$ kPa on the geothermal fluid side and $\Delta P_c \leq 40$ kPa on the district heating water side were cited. In the final analysis, the pressure drop was assessed using the calculated hydraulic behavior of the chosen heat exchanger configuration rather than these values as strict design constraints.

3.2 Design Basis and Temperature at Which Characteristics Are Evaluated

The design heat load is $Q = 2$ MW. The temperatures for district heating water are given as $T_{c,in}=40^\circ\text{C}$ and $T_{c,out}=65^\circ\text{C}$. The inlet temperature for the geothermal brine is $T_{h,in}=90^\circ\text{C}$, and the value of $T_{h,out}$ is initially unknown; this value will be determined by the energy balance in the Design and Calculation section. Fluid properties are evaluated based on the average temperature for district heating water $T_{avg,c}=52.5^\circ\text{C}$ (average of the 40–65°C range). On the brine side, the specific heat has been assumed to be close to that of water, considering the TDS values reported for Balçova waters. [2]

3.3 The Effect of Number of Plates (N_p)

In this work, the number of plates (N_p) was chosen as the key parametric design variable to explore the performance of the plate heat exchanger. The number of plates has a significant impact on the thermal capacity and efficiency of the heat exchanger because it directly influences the total heat transfer area. An increase in the number of plates often results in an increase in heat transfer surface area and achievable heat load.

Changes in the number of plates, however, have an impact not only on the heat transfer area but also on the heat exchanger's hydraulic behaviour. As Np rises, the number of parallel flow channels increases, resulting in a drop-in mass flow rate and average flow velocity through each channel. As a result, changes occur in the Reynolds number, Nusselt number, and convective heat transfer coefficient.

Therefore, the impact of the number of plates on heat exchanger performance cannot be assessed purely on the basis of increased surface area. Although a larger number of plates increases the heat transfer area, diminishing returns may occur beyond a certain point due to decreasing channel velocities and increasing equipment dimensions. Furthermore, increasing the number of plates can have a negative influence on both pressure losses and overall system size.

This study studied the effects of three distinct plate numbers ($Np = 80, 100$, and 120). The computations are only shown in full for the representative instance of $Np = 100$, while the results for $Np = 80$ and $Np = 120$ are summarized in Tables.

4.DESIGN AND CALCULATION

For this project, the design heat load (Q) is given, while the stream temperatures, thermophysical properties, and other required design inputs are obtained from the literature sources specified in the report. The district-heating water temperatures are specified, whereas the geothermal brine outlet temperature is initially unknown and is determined from an energy balance during the sizing calculations. The calculation procedure in this section follows the methodology described in Coulson & Richardson (Sinnott, Vol. 6) as a reference for the design calculations. The number of plates (Np) is used as the primary design parameter to satisfy the required duty.

Assumptions:

- System is at steady-state
- Our system has plate heat exchanger
- Geothermal brine water properties are assumed to be same as district heating water properties.
- Constant properties
- Pipe surface is smooth
- Heat transfer does not exist outside of the system
- Fluid is incompressible
- We neglected R_w in the calculations because it was very small.

4.1. Mass Flow Rate Calculations (Given Heat Duty)

$$\dot{m}_c = \frac{Q}{c_{p,c}(T_{c,out} - T_{c,in})}$$

This equation comes from the cold side energy balance: $Q = \dot{m}_c c_{p,c} \Delta T_c$. Therefore, if Q and the cold-side temperatures are known, the district water flow rate can be directly determined.

$$\dot{m}_c = \frac{2 \times 10^3 \text{ kW}}{4.182 \frac{\text{kJ}}{\text{kg} \cdot \text{K}} (65 - 40) \text{ K}}$$
$$\dot{m}_c = 19.131 \frac{\text{kg}}{\text{s}}$$

Q(kJ/s)	Tc,in(°C)	Tc,out(°C)	Cold Fluid Specific Heat(J/kg*K)	Cold Fluid Mass Flow Rate(kg/s)
2000	40	65	4.182	19.131

Table 1: Parameters for $\dot{m}_c$ calculations

4.2. Energy Balance Calculation

This equation comes from the hot side energy balance. That is, when $\dot{m}_h$ is selected, the geothermal outlet temperature is automatically determined.

$$\text{Assume } \dot{m}_h = 80 \frac{m^3}{h} = 16.383 \text{ kg/s}$$

$$Q = \dot{m}_h * c_{p,h} (T_{h,i} - T_{h,o})$$

$$T_{h,i} - T_{h,o} = \frac{Q}{\dot{m}_h * c_{p,h}}$$

$$(90 - T_{h,o}) = \frac{2 \times 10^3 \text{ kW}}{(16.383 \frac{\text{kg}}{\text{s}} * 4.194 \frac{\text{kJ}}{\text{kg} * \text{K}})}$$

$$T_{h,o} = 60.795 \text{ °C}$$

Q (kJ/s)	Specific Heat (J/kg*K)	Hot Fluid Mass Flow Rate (kg/s)	Th,in(°C)	Th,out(°C)
2000	4.194	16.383	90	60.795

Table 2: Geothermal Brine Outlet Temperatures

4.3. Capacity Ratio Terms: C_{min} , C_{max} , C_r

$$C_{min} = \min(C_h, C_c)$$

$$C_{max} = \max(C_h, C_c)$$

$$C_h = \dot{m}_h * c_{p,h}$$

$$C_c = \dot{m}_c * c_{p,c}$$

$$C_h = 16.383 \frac{\text{kg}}{\text{s}} * 4.194 \frac{\text{kJ}}{\text{kg} * \text{K}} = 68.481 \frac{\text{kJ}}{\text{K} * \text{s}}$$

$$C_c = 19.131 \frac{\text{kg}}{\text{s}} * 4.182 \frac{\text{kJ}}{\text{kg} * \text{K}} = 79.998 \frac{\text{kJ}}{\text{K} * \text{s}}$$

$$C_{min} = C_h = 68.481 \frac{\text{kJ}}{\text{K} * \text{s}}$$

$$C_r = \frac{C_{min}}{C_{max}}$$

$$C_r = 0.856$$

In the ε -NTU method, C_{min} limits the maximum possible heat transfer. Furthermore, the C_r value determines the calculation of the required NTU in the ε -NTU relationship.

Fluid	m (kg/s)	cp (kJ/kg*K)	C (kJ/K*s)
Hot	16.383	4.194	68.481
Cold	19.131	4.182	79.998

Table 3: Ch and Cc Values

4.4. Maximum Possible Heat Transfer and Required Effectiveness

$$Q_{max} = C_{min}(T_{h,in} - T_{c,in})$$

$$Q_{max} = 68.481 \frac{kJ}{s \cdot K} (90-40) K = 3,424.05 \frac{kJ}{s}$$

This provides the theoretical upper limit that can be physically achieved; it is a reference point that shows how difficult the design is.

$$\varepsilon = \frac{Q}{Q_{max}}$$

$$\varepsilon = \frac{2 \cdot 10^3 kW}{3.424.05 kW} = 0.584$$

Effectiveness is the ratio of the targeted duty to the theoretical maximum. The NTU requirement of the design is determined based on this ε .

4.5. ε -NTU Analysis and Target UA

$$NTU = \frac{1}{C_r - 1} \ln \left(\frac{\varepsilon - 1}{\varepsilon C_r - 1} \right)$$

$$NTU = \frac{1}{0.856 - 1} \ln \left(\frac{0.584 - 1}{0.584 * 0.856 - 1} \right)$$

$$NTU = 1.279$$

$$UA = NTU * C_{min}$$

$$UA = 1.279 * 68.481 \frac{kJ}{K \cdot s} = 87.587 \frac{kJ}{K \cdot s}$$

This converts the required NTU into a required conductance UA. The sizing step will ensure the selected plate count provides at least this UA.

4.6. Calculation of Determined UA

4.6.1. Channel Count and Flow Per Channel

$$A_{flow} = N_{ch} * A_c$$

$$A_{flow} = 50 * 0.00087m^2$$

$$A_{flow}=0.0435m^2$$

$$u = \frac{m}{\rho A_{flow}}$$

$$u_c = \frac{19.131 \frac{kg}{s}}{987.128 \frac{kg}{m^3} * 0.0435 m^2}$$

$$u_c = 0.445528 \frac{m}{s}$$

$$u_h = \frac{16 \frac{kg}{s}}{974.537 \frac{kg}{m^3} * 0.0435 m^2}$$

$$u_h = 0.386461 \frac{m}{s}$$

Fluid Type	Np	Nch	u (m/s)	Aflow(m^2)	Ac (m^2)
Hot fluid	80	40	0.48307644	0.0348	0.00087
Cold fluid		40	0.55691		
Hot fluid	100	50	0.38646115	0.0435	
Cold fluid		50	0.445528		
Hot fluid	120	60	0.322051	0.0522	
Cold fluid		60	0.371273		

Table 4: u, Aflow, Ac Values According to the Np Parameter

4.6.2. Reynolds and Prandtl Numbers

$$D_h = \frac{4A_c}{P_w}$$

$$\text{for } Np=100 \quad D_h = \frac{4*0.00087m^2}{0.586m} = 0.005939m$$

$$Re = \frac{\rho \cdot u \cdot D_h}{\mu}$$

$$Re_c = \frac{987.128 \frac{kg}{m^3} \cdot 0.445528 \frac{m}{s} \cdot 0.006}{0.000528 \frac{Ns}{m^2}} = 4946.4784$$

$$Re_h = \frac{974.537 \frac{kg}{m^3} \cdot 0.386461 \frac{m}{s} \cdot 0.004}{0.000372 \frac{Ns}{m^2}} = 6012.9773$$

Reynolds number determines the flow regime and selects the appropriate Nusselt correlation. It should be calculated separately for hot and cold sides using their own properties.

Fluid Type	Np	Nch	Re	Dh (m)
Hot fluid	80	40	7516.22161	0.005939
Cold fluid		40	6183.098	
Hot fluid	100	50	6012.97728	
Cold fluid		50	4946.478	
Hot fluid	120	60	5010.8144	
Cold fluid		60	4122.065	

Table 5: Reynolds Number Calculations

$$Pr = \frac{c_p \cdot \mu}{k}$$

$$Pr_c = \frac{4181.6 \frac{J}{kg} \cdot 0.000528 \frac{kg}{m \cdot s}}{0.645 \frac{W}{m \cdot K}}$$

$$Pr_c = 3.423077$$

$$Pr_h = \frac{4194 \frac{J}{kg} \cdot 0.00037196 \frac{kg}{m \cdot s}}{0.66542 \frac{W}{m \cdot K}}$$

$$Pr_h = 2.344384$$

Fluid Type	cp(J/kg)	μ(kg/m.s)	k(W/m*K)	Pr
Hot fluid	4194	0.00037196	0.66542	2.344384
Cold fluid	4181.6	0.000528	0.645	3.423077

Table 6: Pr Numbers

4.6.3. Convection Coefficient

$$[Nu = C_1 Re^p Pr^{C_3}]$$

$$\text{with } C_1 = 0.2414, p = 0.64, C_3 = 0.4$$

if

$$\left[\begin{array}{c} 700 < Re < 20000 \\ \beta = 45^\circ \end{array} \right]$$

$$Nu = 0.2414Re^{0.64}Pr^{0.4}$$

$$Nu_h = 0.2414 * 6012.9773^{0.64} * 2.344384^{0.4} = 88.99510$$

$$Nu_c = 0.2414 * 4946.4784^{0.64} * 3.423077^{0.4} = 91.38049$$

$$h = \frac{Nu * k}{D_h}$$

$$h_h = \frac{88.99510 * 0.665 \frac{W}{m \cdot K}}{0.005939m} = 9971.955 \frac{W}{m^2 K}$$

$$h_c = \frac{91.38049 * 0.645 \frac{W}{m \cdot K}}{0.005939m} = 9925.024 \frac{W}{m^2 K}$$

Fluid Type	Np	Nch	h(W/m ² *K)	Nu
Hot fluid	80	40	11502.78	102.657
Cold fluid		40	11448.64	105.4086
Hot fluid	100	50	9971.956	88.9951
Cold fluid		50	9925.024	91.38049
Hot fluid	120	60	8873.692	79.19361
Cold fluid		60	8831.929	81.31627

Table 7: Convection Coefficient and Nusselt Number Calculations

4.6.4. Overall Heat-Transfer Coefficient U

$$U = \left(\frac{1}{h_h} + R_{f,h} + R_w + R_{f,c} + \frac{1}{h_c} \right)^{-1}$$

This combines convection on both sides, fouling resistances, and wall resistance into a single overall coefficient. Since fouling is included in your design, $R_{f,h}$ and $R_{f,c}$ must be explicitly accounted for.

Side	Fluid Type	Fouling Factor (R_f) (m ² ·K/W)	Reference
Hot side	Towns water (hard)	0.00017	Table 12.9
Cold side	Cooling water (treated)	0.00012	Table 12.9

Table 8: Fouling Factor According to Fluid Type

$$U = \left(\frac{1}{9971.955} + 0.00017 + 0.00012 + \frac{1}{9925.024} \right)^{-1}$$

$$U = 2036.508 \frac{W}{m^2 \cdot K}$$

Fluid Type	Np	Nch	A(m ²)	h(W/m ² *K)	U(W/(m ² *K))
Hot fluid	80	40	34.8	11502.7781	2153.863
Cold fluid		40		11448.64	
Hot fluid	100	50	43.5	9971.95556	2036.508
Cold fluid		50		9925.024	
Hot fluid	120	60	52.2	8873.69172	1938.292
Cold fluid		60		8831.929	

Table 9: Value of Area and Overall Convection Coefficient

4.6.5. Area Calculation for Each N_p Value

$$A = N_p A_{plate}$$

$$A = 100 * 0.435 m^2$$

$$A = 43.5 m^2$$

Fluid Type	Np	Nch	A(m^2)	Plate length (m)	Aplate(m^2)
Hot fluid	80	40	34.8	1.5	0.435
Cold fluid		40			
Hot fluid	100	50	43.5		
Cold fluid		50			
Hot fluid	120	60	52.2		
Cold fluid		60			

Table 10: Aplate Value

4.6.6. Overall Conductance Calculation UA for Np Value

$$UA = U * A$$

$$UA = 2036.508 \frac{W}{m^2 * K} * 43.5 m^2$$

$$UA(100) = 87587 \frac{W}{K}$$

Fluid Type	Np	Nch	A(m ²)	U(W/(m ² *K))	UA(W/K)
Hot fluid	80	40	34.8	2153.863	74954.4324
Cold fluid		40			
Hot fluid	100	50	43.5	2036.508	88588.098
Cold fluid		50			
Hot fluid	120	60	52.2	1938.292	101178.8424
Cold fluid		60			

Table 11: UA Calculations

4.7. Pressure Drop Calculations

$$\Delta P_p = 8j_f \left(\frac{L_p}{D_h} \right) \left(\frac{\rho u^2}{2} \right)$$

$$L_p = L * \text{pass number}$$

$$j_f = 0.6 * Re^{-0.3}$$

$$\Delta P_{pt} = 1.3 \left(\frac{\rho u_{pt}^2}{2} \right) * N_{pass}$$

$$u_{pt} = \frac{\dot{m}}{\rho * A_p}$$

$$A_p = \frac{\pi D_{pt}^2}{4}$$

$$u_{pt} = \frac{4\dot{m}}{\rho \pi D_{pt}^2}$$

$$\Delta P_{total} = \Delta P_p + \Delta P_{pt} \quad \text{for each fluid side}$$

For Np=100

For cold water:

$$Re=4946.478 \quad \dot{m} = 19.131 \frac{kg}{s} \quad D_h = 0.005939$$

$$\rho = 987.128 \frac{kg}{m^3}$$

$$u = 0.445528 \text{ m/s}$$

Assume $D_{pt} = 0.1$ [4]

$$L=1.5m$$

$$Lp = 1.5m * 1$$

$$L_p = 1.5m$$

$$j_f = 0.6 * (4946.478)^{-0.3}$$

$$j_f = 0.046759$$

$$u_{pt} = \frac{4 * 19.131 \frac{kg}{s}}{987.128 \frac{kg}{m^3} * \pi * (0.1m)^2}$$

$$u_{pt} = 2.4676 \frac{m}{s}$$

$$\Delta P_p = 8 * 0.046759 \left(\frac{1.5m}{0.005939m} \right) \left(\frac{987.128 \frac{kg}{m^3} * (0.445528 \frac{m}{s})^2}{2} \right)$$

$$\Delta P_p = 9256.067 \frac{kg}{m * s^2} = 9256.067 Pa$$

$$\Delta P_{pt} = 1.3 \left(\frac{987.128 \frac{kg}{m^3} * (2.4676 \frac{m}{s})^2}{2} \right) * 1$$

$$\Delta P_{pt} = 3906.936 \frac{kg}{m * s^2} = 3906.936 Pa$$

$$\Delta P_{total} = 9256.067 Pa + 3906.936 Pa$$

$$\Delta P_{total} = 13163.006 Pa$$

For hot (brine) water:

$$Re=6012.97728 \quad \dot{m} = 16.383 \frac{kg}{s} \quad D_h = 0.005939$$

$$\rho = 974.537 \frac{kg}{m^3}$$

$$u = 0.3865 m/s$$

Assume $D_{pt} = 0.1$ [4]

$$L=1.5m$$

$$Lp = 1.5m * 1$$

$$L_p = 1.5m$$

$$j_f = 0.6 * (6012.97728)^{-0.3}$$

$$j_f = 0.0441$$

$$u_{pt} = \frac{4 * 16.383 \frac{kg}{s}}{974.537 \frac{kg}{m^3} * \pi * (0.1m)^2}$$

$$u_{pt} = 2.1405 \frac{m}{s}$$

$$\Delta P_p = 8 * 0.0441 \left(\frac{1.5m}{0.005939m} \right) \left(\frac{974.537 \frac{kg}{m^3} * (0.3865 \frac{m}{s})^2}{2} \right)$$

$$\Delta P_p = 6485.954 \frac{kg}{m * s^2} = 6485.954 Pa$$

$$\Delta P_{pt} = 1.3 \left(\frac{974.537 \frac{kg}{m^3} * (2.1405 \frac{m}{s})^2}{2} \right) * 1$$

$$\Delta P_{pt} = 2902.165 \frac{kg}{m * s^2} = 2902.165 Pa$$

$$\Delta P_{total} = 6485.954 Pa + 2902.165 Pa$$

$$\Delta P_{total} = 9388.119 Pa$$

Fluid Type	Np	Nch	Re	Dh (m)	u (m/s)	ρ (kg/m ³)	Plate length (m)	jf	u_pt(m/s)	ΔP_p (Pa)	ΔP_{pt} (Pa)	ΔP_{total} (Pa)
Hot fluid	80	40	7516.222	0.00593857	0.483076	974.537	1.5	0.041243	2.140451	9476.567	2902.165	12378.73
Cold fluid			6183.098		0.55691	987.128		0.043731	2.467597	13527.03	3906.929	17433.96
Hot fluid	100	50	6012.977		0.386461	974.537		0.0441	2.1405	6485.954	2902.165	9388.119
Cold fluid			4946.478		0.445528	987.128		0.046759	2.4676	9256.0697	3906.936	13163.006
Hot fluid	120	60	5010.814		0.322051	974.537		0.046578	2.140451	4756.592	2902.165	7658.757
Cold fluid			4122.065		0.371273	987.128		0.049387	2.467597	6789.648	3906.929	10696.58

Table 12: Pressure Drop Calculations

5. FINAL DESIGN

The final configuration of the plate heat exchanger has been established as a consequence of the thermal and hydraulic analyses carried out. The chosen design is a single-pass (1–1), counter-flow plate heat exchanger that is set up to let the district heating water and geothermal fluid pass through different channels. In the final design, the geothermal fluid's outlet temperature is determined using the energy balance and NTU-efficiency method, while the district heating water's inlet and outlet temperatures are kept at 40 °C and 65 °C, respectively.

The internal flow pattern of the chosen final design is schematically depicted in Figure 1. The hot and cold fluid flow directions, the inlet and outlet temperatures, and basic geometric parameters like plate length (L) and channel spacing (s) are shown in the figure. The purpose of this schematic representation is to give a visual understanding of the geometric assumptions and flow pattern utilized in the design calculations.

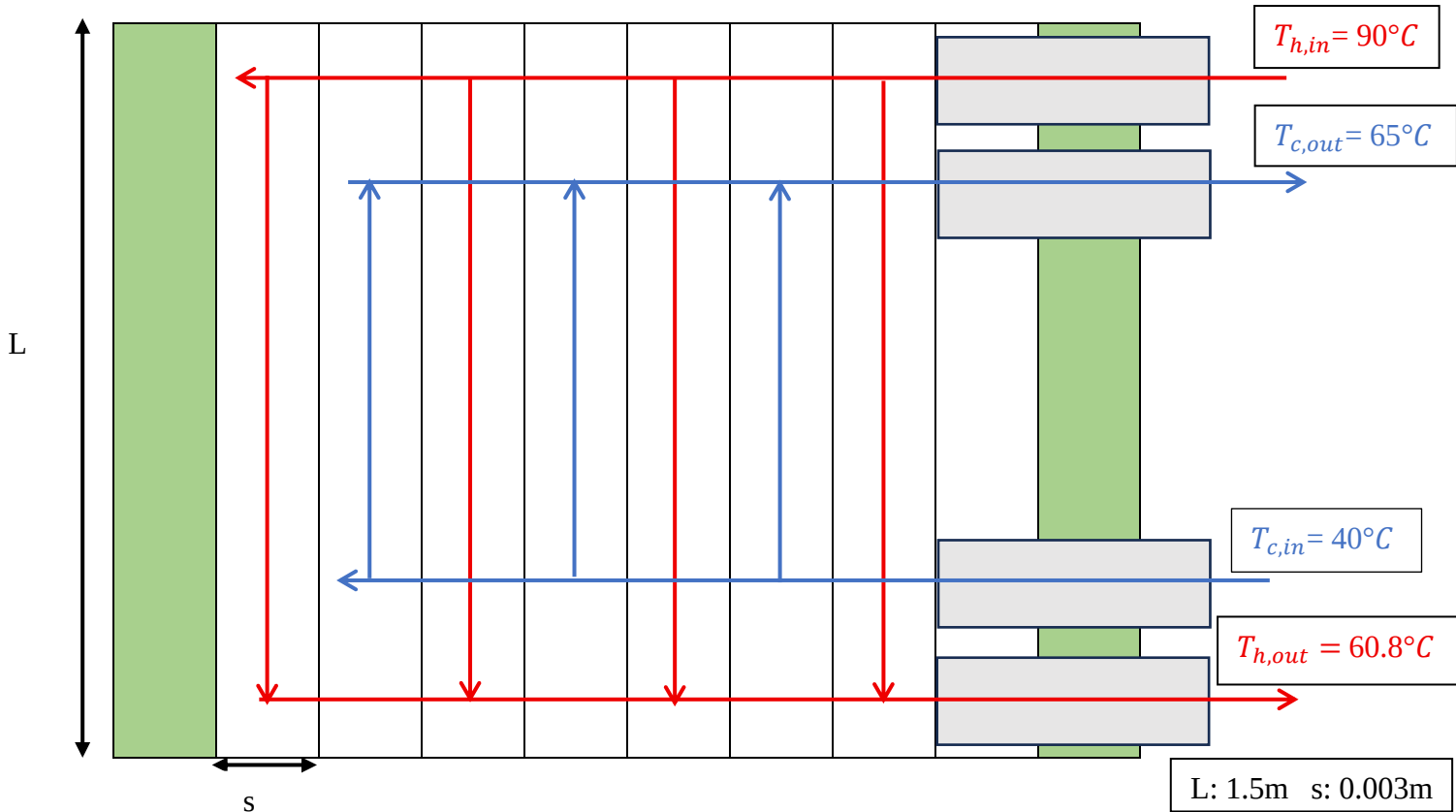


Figure 1: Schematic representation of the internal flow arrangement of the counter-current single-pass plate heat exchanger used in the geothermal district heating system

$$\dot{m}_h = 16 \text{ kg/s} \quad \dot{m}_c = 19.131 \text{ kg/s} \quad \text{Number of plates } (N_p) = 100 \quad Q = 2 \text{ MW}$$

6. RESULTS AND DISCUSSION

This paper intends to design a heat exchanger transferring 2 MW of thermal energy from a low, temperature geothermal source for a district heating water system. The district heating water was kept at a constant temperature of 40°C (inlet) and 65°C (outlet), while the geothermal fluid temperature at the inlet was considered 90°C. Since no direct reference was made to the temperature of the geothermal fluid at the outlet, it was first derived by an energy balance on the hot side and then by selecting a mass flow rate of the hot fluid from the literature. By setting the mass flow rate of the hot side $\dot{m}_h=16.383$ kg/s, the temperature of the geothermal fluid at the outlet was $T_{h,out}=60.8^\circ\text{C}$. Hence, it is evident from this result that the heat transfer under counterflow operation is feasible since the temperature of the geothermal fluid outlet is very close to the temperature of the district water being heated.

The heat capacity ratios obtained are $C_h=68.481$ kJ/Ks for the hot side and $C_c=79.998$ kJ/Ks for the cold side, and the minimum heat capacity ratio $C_{min}=68.481$ kJ/Ks was obtained. This parameter carries significant importance as it essentially sets the limit to the maximum theoretical heat transfer rate the system is capable of. As a result, the maximum attainable heat transfer rate $Q_{max}=3424$ kW was determined and the corresponding heat exchanger efficiency $\varepsilon=Q/Q_{max}=0.584$ was obtained.

The calculated effectiveness indicates that the required heat transfer rate can be achieved without operating near the theoretical limit, while still satisfying the necessary overall heat transfer conductance.

From the NTU heat exchanger characterization method, the heat exchanger size, dependency parameter in terms of transfer units was established as $\text{NTU}=1.279$, which is equivalent to an overall heat transfer conductivity target of $\text{UA}=87.6$ kW/K.

Such a UA value stands as the cornerstone thermal design goal that would allow the accomplishment of a heat transfer rate of 2 MW under the given inlet and outlet conditions with a high level of confidence.

In this case, a gasketed plate heat exchanger was the equipment that was chosen. This is a very appropriate choice as it is easy to maintain and clean and is capable of providing high heat transfer coefficients in a liquid, to, liquid service, unlike the situations of fouling and scaling which are usually the problems encountered in a geothermal context. Moreover, the implementation of a two, loop design (geothermal brine, clean water) excludes the mineralized

geothermal fluid from the district heating network, thereby raising the general reliability of the system. A counterflow, single, pass (1, 1) configuration was chosen to ensure the temperature driving force is more effectively exploited and the heat transfer performance is maximized.

The number of plates N_p was chosen as the primary design parameter in the parametric study, and values of $N_p=80, 100, 120$ were evaluated. This is because, by altering the number of parallel channels, N_p not only modifies the overall area but also the flow rate, velocity, and ultimately the Reynolds number per channel. The findings supported this effect: the number of channels rose, and the average channel velocity reduced as N_p grew (e.g., velocity on the hot side was $0.483 \text{ m/s} \rightarrow 0.386 \text{ m/s} \rightarrow 0.322 \text{ m/s}$). The Nusselt number and convection coefficients fell concurrently with the Reynolds numbers (hot side: $7516 \rightarrow 6013 \rightarrow 5011$, cold side: $6183 \rightarrow 4946 \rightarrow 4122$).

However, the total conductivity UA typically rose since the area growth was prominent in the studied range. $U \approx 2154 \text{ W/m}^2\cdot\text{K}$, $A \approx 34.8 \text{ m}^2$, and $UA \approx 74.95 \text{ kW/K}$ were determined for $N_p = 80$; since this value is less than the aim of 87.6 kW/K , it is insufficient to ensure 2 MW under the same conditions. $U \approx 2036 \text{ W/m}^2\cdot\text{K}$, $A \approx 43.5 \text{ m}^2$, and $UA \approx 88.6 \text{ kW/K}$ were computed for $N_p = 100$, closely matching the required UA . $U \approx 1938 \text{ W/m}^2\cdot\text{K}$, $A \approx 52.2 \text{ m}^2$, and $UA \approx 101.2 \text{ kW/K}$ were achieved for $N_p=120$; it necessitated more plates and larger equipment. Because it has the ideal number of plates that reach the goal, $N_p=100$ stands out as the most balanced alternative among those tested.

Accounting for the fouling impact in geothermal applications increases the realism of the design. The fouling resistances on both sides are included in this report; this method reduces U in comparison to the ideal clean surface condition and raises the possibility of reaching the 2 MW target even in the event of surface contamination or mineral accumulation that may develop over time.

This decision is also supported by the pressure drop calculation. The Reynolds number and channel velocity on the cold water side were used to get $\Delta P_c=13163\text{Pa} \approx 13.16\text{kPa}$ for $N_p=100$. This result indicates a reasonable operation in terms of pumping power and is much below the 40 kPa limit stated in the design limitations. The pressure drop is similarly observed to be much below the 50 kPa limit (of the same order, $\sim 10 \text{ kPa}$ levels) when the same technique and channel velocity $N_p=100$ are applied for the hot (geothermal) side. Thus, the updated hydraulic control shows that the $n_p=100$ selection is both thermally and hydraulically practical.

Geothermal district heating has definite advantages over fossil fuel-based heating in terms of sustainability and environmental impact: it uses a local or renewable heat source and reduces emissions associated with combustion. By separating the mineralized geothermal fluid from the distribution line, the two-pass PHE method lowers the chance of corrosion and leaks. It also lessens the need for maintenance and chemicals, which lessens the environmental impact. However, scale control, minimizing auxiliary electricity consumption (pumping), and environmentally sound management of geothermal brine (typically reinjection) are crucial; consequently, system sustainability is directly related to a conservative design approach that takes ΔP limits and fouling into account.

7. CONCLUSION

The ϵ -NTU approach was used in this project to size a plate heat exchanger that could supply the required heat load of $Q = 2$ MW. [4] The main design parameter that was assessed was the impact of the number of plates (N_p) on performance. Based on the energy balance, the temperature of the geothermal brine outlet was found to be approximately 60.8°C; the capacity ratios were used to determine the necessary UA (target) = 87.6 kW/K to complete the target task.

The parametric findings demonstrated the anticipated trend of a rise in area as N_p increases, but a slow decline in U as a result of the flow being split into numerous channels. However, the overall performance metric $UA = U \cdot A$ increased with N_p ; for $N_p=80$, UA stayed below the target, indicating that 2 MW could not be accomplished; for $N_p=100$, UA exceeded the target and the task was completed; and for $N_p=120$, the task was completed with a higher confidence margin but required larger equipment. As a result, $N_p=100$ is the best option among those considered because it uses the fewest plates to achieve the desired heat load.

For $N_p=100$, $\Delta P \approx 13.16$ kPa was determined on the district heating water side in the hydraulic analysis. This value remained below the 40 kPa limit specified in the project requirements, indicating that the design is suitable in terms of pumping power.[4]

As a result, the selected approach (ϵ -NTU + N_p parametric screening) has yielded a feasible PHE design that meets the 2 MW target from both thermal and hydraulic perspectives. In future studies, the assumption that brine properties are ‘close to water’ [2] and the effect of correlation selections on the results [3] can be examined in more detail with sensitivity analysis; additionally, verifying the U and ΔP values with manufacturer data will make the design more reliable.

Additionally, the plate-type heat exchanger design developed in this study is not only technically feasible but also offers significant advantages in terms of sustainable energy consumption. Separating the mineral-rich geothermal fluid from the distribution line extends the system's lifespan, thereby reducing corrosion, contamination, and maintenance requirements.[2] Furthermore, the design minimises pumping energy and reduces auxiliary energy usage while operating with tolerable pressure drops.[4] In this way, the proposed design offers a sustainable and energy-efficient solution for geothermal district heating applications.

PROJECT PLANNING

Table 13: Gantt Chart

		2.01.2026	3.01.2026	4.01.2026	5.01.2026	6.01.2026	7.01.2026	8.01.2026	9.01.2026	10.01.2026
Abstract and Introduction	All members									
Theory	Sena Uçur Esra Nur Yılmaz									
Heat Exchanger Configuration & Design Justification	Sena Uçur Esra Nur Yılmaz									
Parametric Study-Parameter Selection	Sena Uçur Esra Nur Yılmaz									
Design Calculations (NTU Method)	Fatma Ceren Uygur Secem Karakaya									
Parametric Study-Calculations & Results	Fatma Ceren Uygur Secem Karakaya									
Final Heat Exchanger Design	Esra Nur Yılmaz									
Results and Discussion	All members									
Conclusion	All members									
Final Review and Submission	All members									

REFERENCES

- [1] Bergman, T. L., Lavine, A. S., Incropera, F. P., & DeWitt, D. P. (2011). *Fundamentals of heat and mass transfer* (7th ed.). John Wiley & Sons.
- [2] Polat, C. (2010). *Numerical modeling of Balçova geothermal field* [Master's thesis, Middle East Technical University].
- [3] Pradhan, R. L., Ravikumar, D., & Pradhan, D. L. (2019). Review of Nusselt number correlation for single phase fluid flow through a plate heat exchanger to develop C# code application software. *IOSR Journal of Mechanical and Civil Engineering*, 1–8. <https://www.iosrjournals.org/iosr-jmce/papers/RDME-Volume4/RDME-30.pdf>
- [4] Sinnott, R. K. (2005). *Coulson & Richardson's chemical engineering: Volume 6 – Chemical engineering design* (4th ed.). Elsevier Butterworth-Heinemann.